

Input image

HGNetv2-B4
Hybrid encoder

Decoder layers 1–5

Features F

Side probabilities

Deformable cross-attention + gateway

Training only
Exterior mean m

$F' = F - 0.5m$

Replayed BCSA
content and Δ

FFN + heads

Auxiliary loss

BCSA
Expected region state

State correction

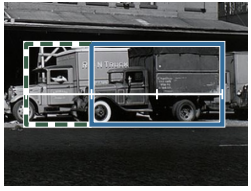
q

q^+

FFN + heads

(a) From candidate rectangles to a query update

Candidate rectangles



-- **A** wide — **B** tight

Interval states

Cell write	4	2	3	Box prob.
A [1, 3]	4 →	4 →	5	× ¼
B [2, 3]	–	2 →	4	× ¾

$h_j = \frac{1}{2} h_{j-1} + w_j$ reset at each left edge

Boundary-weighted read

$\bar{H}_q = \frac{1}{4}(5) + \frac{3}{4}(4) = 4.25$

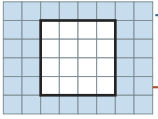
q Observed query

$\Sigma \rightarrow \Phi(q, \bar{H}_q) \rightarrow (+) \rightarrow q^+$

Project the state and add it to the query.

(b) Training only: decoder replay from the shared encoder features

Encoder grid



Shaded: box exterior

Encoder memory F

$F' = F - 0.5m$

m: exterior mean

Reuse the query q and box probabilities π_q

Content v', u' → $w' = \Delta' v' u'$ write

Step size Δ' → $a' = e^{-\Delta' e^{\theta}}$ retain

Replayed **BCSA** → FFN, heads → $0.1\,\mathcal{L}_{\text{aux}}$

Training targets